BHAGWANT SINGH

Assistant Professor
Department of Computer Science and Engineering
Shivalik University, Dehradun
PhD Scholar (Computer Science and Engineering)
Punjabi University, Patiala

Research Areas: AI Security • Malware Analysis • Explainable Artificial Intelligence (XAI) • Adversarial Machine Learning • Cybersecurity

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Research Profile

My research focuses on the application of Artificial Intelligence to Cybersecurity, with particular emphasis on AI-driven malware detection, explainable artificial intelligence (XAI), adversarial machine learning, and intelligent threat analytics. My current Ph.D. research investigates hybrid static-dynamic malware analysis frameworks that integrate machine learning with explainability techniques to improve the transparency, robustness, and reliability of malware detection systems. In addition to cybersecurity research, I contributed to the MeitY-funded BHASHINI initiative as a Senior Computer Scientist, where I led the technical development of transformer-based multilingual NLP frameworks, dataset engineering, model training, and deployment for low-resource Indian languages. My long-term research vision is to establish trustworthy AI-driven cybersecurity systems by integrating explainable machine learning, malware intelligence, adversarial robustness, and secure AI techniques to address emerging cyber threats while translating research into scalable tools for academia, industry, and government.

Research Vision

My recent research focuses on the development of machine learning frameworks for **malware analysis and cybersecurity analytics**, with emphasis on hybrid static–dynamic feature modeling and **Explainable Artificial Intelligence (XAI) techniques**. This work investigates how interpretable machine learning models can improve transparency, robustness, and reliability in real-world security systems. In parallel, I contribute to multilingual NLP research through the **MeitY-funded BHASHINI initiative**. This work involves developing multi-task transformer-based shallow parsing models, preprocessing annotated multilingual datasets, and training transformer-based models using **XLM-R** and **MURIL**. My future research aims to develop **scalable AI-driven cybersecurity systems** capable of detecting evolving malware threats and adversarial attacks. Planned work includes developing interpretable deep learning models for malware analysis, studying adversarial robustness in machine learning systems, and exploring multilingual NLP techniques for security intelligence and threat analysis. Future research will pursue funding opportunities from national research programs supporting **AI, cybersecurity, and language technologies**, including initiatives supported by **MeitY, SERB, and DST**.

Research Highlights

- Total Publications: 17
- Google Scholar Citations: 39
- h-index: 4
- **Patent:** Registered Industrial Design: 1 (Government of India)
- Articles Under Review: 4

Research Impact

- Developed multilingual NLP resources and AI pipelines supporting more than 10 low-resource Indian languages under the MeitY BHASHINI initiative.
- Built transformer-based shallow parsing models achieving task-dependent F1-scores up to 92%.
- Designed and implemented multilingual data preprocessing pipelines and prepared model-ready datasets for transformer-based language technology research.
- Deployed trained models using a FastAPI-based inference service for internal evaluation and demonstration of the multilingual NLP pipeline.

Funded Research Participation

Duration: November 2024 – March 2026

HIMANGY - Indian Language to Indian Machine Translation (BHASHINI Initiative)
Funding Agency: Ministry of Electronics & Information Technology (MeitY), Government of India
Role: Senior Computer Scientist (Project Scientist)
This project involved collaboration with research teams from **IIIT Hyderabad**, **National Institute of Technology Patna**, **CDAC Delhi**, **CDAC Noida**, **CDAC Bangalore**, **IIIT Bhubaneswar**, **MIT**, **Government College for Women Jammu**, **University of Kashmir**, **DAIICT**, **Indian Institute of Technology Patna** for multilingual dataset development and language technology research under the BHASHINI initiative.

Research Interests

- AI-Based Malware Detection
- Adversarial Machine Learning
- Explainable Artificial Intelligence (XAI) in Cybersecurity
- Network Intrusion Detection Systems
- Transformer-Based NLP for Security Analytics

EDUCATION

Doctor of Philosophy October 2023 – Present
Punjabi University, Patiala
Department of Computer Science and Engineering
Ph.D. in Computer Science and Engineering (Expected May 2027)
Research Area: AI-driven malware detection using hybrid static-dynamic analysis and explainable machine learning techniques.

Master of Technology August 2020 – June 2022
Punjabi University, Patiala
Department of Computer Science and Engineering
CGPA: 9.53
Focus: Cybersecurity, Malware Analysis and Network Security technologies.

Bachelor of Technology August 2015 – June 2019
Rayat Bahra University
Department of Computer Science and Engineering
Overall CGPA: 8.15
Research Training Distinction: 10.0 CGPA

EMPLOYMENT HISTORY

Assistant Professor May 2026 – Present
Shivalik University, (Dehradun, Uttarakhand, India)
Department of Computer Science and Engineering

- Leading the drafting and development of the curriculum and syllabus for the B.Tech CSE Specialization in Cybersecurity under the Centre of Excellence (CoE) in Cybersecurity, aligning the programme with emerging technologies, industry requirements, and academic standards.

Senior Computer Scientist – MeitY BHASHINI (HIMANGY) Project Nov 2024 – March 2026
Research Centre for Technical Development of Punjabi Language, Literature and Culture
Punjabi University, Patiala

- Developed multi-task transformer-based shallow parsing models using **XLM-RoBERTa** and **MURIL** architectures for **10+ low-resource Indian languages** under the MeitY-funded **BHASHINI initiative**.
- Curated and annotated ∼**18,000 sentences** across multiple datasets in **SSF** and **CoNLL** formats supporting tasks such as **POS tagging, chunking, root identification, category classification, morphological analysis, named entity recognition (NER), TAM extraction, and Vibhakti detection**.
- Implemented preprocessing pipelines and trained multi-task transformer models achieving **F1-scores ranging from 80–92% depending on task complexity and language resource availability**.

- Deployed trained models on an internal server environment for demonstration and evaluation of the multilingual NLP pipeline.
- Collaborated with research teams and language experts from **IIIT Hyderabad**, **National Institute of Technology Patna**, **CDAC Delhi**, **CDAC Noida**, **CDAC Bangalore**, **IIIT Bhubaneswar**, **MIT**, **Government College for Women Jammu**, **University of Kashmir**, **DAIICT**, **Indian Institute of Technology Patna**, and **Punjabi University** for multilingual dataset preparation and linguistic resource development under the BHASHINI initiative.

Assistant Professor August 2022 – October 2024
Universal College Patran, Punjab
Department of Computer Science
Delivered undergraduate and postgraduate courses in Network Security and Cybersecurity. Integrated applied machine learning concepts into cybersecurity curriculum delivery. Mentored students in academic projects related to AI and network defense mechanisms.
Courses Taught:

- Computer Networks
- Artificial Intelligence
- Machine Learning
- Network Security
- Deep Learning

Supervised student projects, developed laboratory exercises, and supported curriculum delivery for cybersecurity and AI-related courses.

Research Assistant August 2019 – July 2020
FlipFlop InfoSec Labs
Cybersecurity Wing
Assisted in enterprise-level penetration testing and vulnerability assessment assignments. Supported firewall and intrusion detection system configuration and performance tuning. Conducted security audits to identify network vulnerabilities and mitigation strategies. Contributed to cybersecurity awareness training programs and technical workshops. Gained applied experience in practical network defense and threat assessment methodologies.

Research Projects

1. **AI-Based Malware Detection Framework:**
   Developing hybrid static–dynamic machine learning models for malware classification integrating Explainable Artificial Intelligence (XAI) techniques to improve model interpretability and robustness. Experimental evaluation is conducted using publicly available malware datasets including **EMBER2018, EMBER2024, CIC Malware Dataset, VirusShare, and MalwareBazaar** to analyze model generalization and detection performance.

2. **Multilingual NLP Shallow Parsing System:**
   Designed and implemented transformer-based multi-task models (**XLM-R, MURIL**) for shallow parsing of **10+ low-resource Indian languages** under the **MeitY-funded BHASHINI initiative**. The work involves dataset preprocessing, annotation alignment, and training models for linguistic tasks such as POS tagging, chunking, morphological analysis, and named entity recognition. This work also involved collaboration with research teams from **IIIT Hyderabad**, **National Institute of Technology Patna**, **CDAC Delhi**, **CDAC Noida**, **CDAC Bangalore**, **IIIT Bhubaneswar**, **MIT**, **Government College for Women Jammu**, **University of Kashmir**, **DAIICT**, **Indian Institute of Technology Patna**, and **Punjabi University** for multilingual dataset preparation and language resource development.

PUBLICATIONS

Journal Publications

1. Singh, B., & Cheema, S. S. (2024). Emerging Trends in AI-Powered Malware Detection: A Review of Real-Time and Adversarially Resilient Techniques. *Tuijin Jishu / Journal of Propulsion Technology*, 45(4), 1841–1869. (Scopus Indexed)
2. Singh, B., & Cheema, S. S. (2024). Psychology in Cybersecurity: Unveiling the Human Dimensions of Digital Resilience. *International Journal of Advanced Networking and Applications*, 16(1), 6281–6290. https://doi.org/10.35444/IJANA.2024.16107
3. Singh, B., & Cheema, S. S. (2024). Psychological Defenses in Cyberspace: Unveiling the Significance of Cognitive Shields in Cybersecurity. *International Journal of Advanced Networking and Applications*, 16(1), 6291–6295. https://doi.org/10.35444/IJANA.2024.16108

4. Singh, B., & Cheema, S. S. (2025). A Disruptive Ransomware: Its Impersonation Process, Identification and Avoidance. *International Journal of Multidisciplinary Research and Growth Evaluation*, 6, 238–245.

International Conference Publications

1. Singh, B., & Cheema, S. S. (2023). Next Generation Firewall and Self Authentication for Network Security. In *Proceedings of the 7th International Conference on Image Information Processing (ICIIP 2023)*, Solan, India, pp. 707–713. IEEE. https://doi.org/10.1109/ICIIP61524.2023.10537758
2. Singh, B., Sidhu, K. S., & Nigam, S. (2025). Artificial Intelligence-Based Dataset Optimization to Strengthen Malware Classification Frameworks. In *Proceedings of the 8th International Conference on Image Information Processing (ICIIP 2025)*, Solan (Near Shimla), Himachal Pradesh, India, pp. 107–111. IEEE. https://doi.org/10.1109/ICIIP68302.2025.11346139
3. B. Singh and S. S. Cheema, "Harnessing Artificial Intelligence for Data Preprocessing to Improve Malware Detection Accuracy," 2025 *2nd Global AI Summit - International Conference on Artificial Intelligence and Emerging Technology (AI Summit), Noida, India, 2025*, pp. 1156–1161, doi: 10.1109/AISummit66170.2025.11411213.
4. Singh, B. (2025). Optimizing Malware Detection Through Systematic Data Preprocessing Techniques for Robust Machine Learning Models *Springer Nature Conference Proceedings.* (Accepted)
5. Singh, B. (2025). AI-Based Malware Detection Frameworks: A Comprehensive Review of Adaptive and Explainable Techniques Analysis. *Scopus Indexed Conference Proceedings.* (Accepted)
6. Singh, B. (2026). Two-Phase Framework for Lemma Identification in Low-Resource Languages *Conference Proceeding (NITJ).* (Accepted)

Book Chapters

1. Cheema, S. S., Chand, L., & Singh, B. (2023). Artificial Intelligence and Machine Learning in Factories of the Future: Technological Advancements in the Manufacturing Industry. In *Factories of the Future: Technological Advancements in the Manufacturing Industry.* Scrivener Publishing (Wiley).
2. Singh, B., & Cheema, S. S. (2022). Role of Robots in the Smart Agriculture System. In *Role of Artificial Intelligence and Automation in Modern Age Agriculture* (Vol. 1). Gurmehar Publication. ISBN: 978-93-93579-38-6.
3. Singh, B., & Savia (2022). Machine Learning in Modern Agriculture. In *Role of Artificial Intelligence and Automation in Modern Age Agriculture* (Vol. 1). Gurmehar Publication. ISBN: 978-93-93579-38-6.
4. Singh, B., & Cheema, S. S. (2023). Drones in Agriculture: A Comprehensive Review. In *Role of Artificial Intelligence and Automation in Modern Age Agriculture* (Vol. 2). Gurmehar Publication. ISBN: 978-93-93579-75-1.

Early Publications

1. Singh, B., et al. (2022). Methods for Restricting Unwanted Access on the Networks Using Firewall. *International Journal of Advanced Research in Innovative Discoveries in Engineering and Applications (IJARIDEA)*, 7(2), 40–47.
2. Singh, B., et al. (2022). Network Security Attacks & Techniques to Secure the Networks. *International Journal of Advanced Research in Innovative Discoveries in Engineering and Applications (IJARIDEA)*, 7(2), 40–47.
3. Singh, B., & Cheema, S. S. (2024). Cybersecurity Threats and Mitigation Strategies in Agriculture 4.0 and 5.0: Challenges and Solutions in the Digital Transformation of Agriculture. *JPAS*, 24, 15–32.

Honors and Awards

- Best Paper Award, 5th International Conference on Smart Systems: Innovations in Computing (2025), for the paper titled Optimizing Malware Detection: Effective Pre-Processing Techniques for Enhanced Dataset Preparation.

Technical Expertise

- **Programming:** Python, Bash, PowerShell
- **Machine Learning:** PyTorch, Hugging Face Transformers
- **Cybersecurity:** Metasploit, Burp Suite, Nmap, Wireshark, Tcpdump
- **Deployment & Platforms:** FastAPI, Flask, Linux, Git, CUDA
- **Systems & Networking:** Linux server administration, DNS, DHCP, Firewall Configuration

SERVICE & PROFESSIONAL DEVELOPMENT

Editorial and Peer Review Activities

- **Reviewer, Journal of Information Security and Applications (Elsevier)**, April 2026 – Present
  Evaluating manuscripts in cybersecurity, AI-driven security systems, and malware detection.
- **Reviewer, Current Psychology (Springer Nature)**, January 2026 – Present
  Reviewing interdisciplinary research at the intersection of psychology and cybersecurity.

Department Service

- Served as an Organizing Committee Member for the National Level Seminar on "Role of Robots in the Smart Agriculture System" under the theme Role of Artificial Intelligence and Automation in Modern Age Agriculture (2022–2025). Contributed to seminar coordination, speaker communication, and technical session organization.
- Participated in the development and maintenance of seminar webpages, including event announcements and technical updates.

University Service

- Served as Senior Computer Scientist under the **MeitY-funded BHASHINI (HIMANGY)** initiative, supporting multilingual language technology development, dataset engineering, and project implementation.
- Coordinated with language experts on structured dataset formats, preprocessing, implementation, and documentation of project milestones aligned with government-funded research objectives.

Scholarly Contributions

- Published peer-reviewed interdisciplinary research in cybersecurity, cyber psychology, and AI-driven smart agriculture systems.

Professional Development

- Participated in the Three-Day Research and Skill Development Programme (RSDP) on AI in Cybersecurity and Forensic Science organized by National Forensic Sciences University (NFSU), Goa Campus, Ministry of Home Affairs, Government of India (September 2025).
- Engaged in hands-on technical discussions on applied machine learning models in cyber threat detection and forensic science applications.

Invited Talks and Workshops

- Expert Speaker – Cybersecurity Workshop, Sri Venkateswara College of Engineering, Tirupati
- Technical Expert – National Workshop on Cybersecurity NIT Meghalaya